Finding the no of even/odd/zero elements column wise:

```
TC
File   Edit   Run   Compile   Project   Options   Debug   Break/watch
     Line 17     Col 33  Insert Indent Tab Fill Unindent * E:2PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,nc, r,c,e,o,z;
clrscr();
printf("Enter no of rows and columns ");scanf("%d%d",&nr,&nc);
printf("Enter %d elements\n",nr*nc);
for(r=0;r<nr;r++)for(c=0;c<nc;c++)scanf("%d",&a[r][c]);
puts("\t Even\tOdd\tZero");
puts("--------------------------------------");
for(c=0;c<nc;c++)
{
for(e=o=z=r=0;r<nr;r++)
{
if(a[r][c]==0)z++; else if(a[r][c]%2==0)e++;else o++;}
printf("%d-col \t%d\t%d\t%d\n",c+1,e,o,z);
}
getch();
}
```

```
TC
Enter no of rows and columns 3 3
Enter 9 elements
6 2 8
4 5 9
0 7 0
         Even    Odd     Zero
--------------------------------------
1-col    2       0       1
2-col    1       2       0
3-col    1       1       1
```

```
for( c=0; c<3; c++ )
{
for( r=e=o=z=0; r<3; r++ )
{
if(a[r][c]==0) z++; else if(a[r][c]%2==0)e++; else o++;
}
p("%d-col\t%d\t%d\t%d\n",c+1, e,o,z);
}

}
```

6 0,0	2 0,1	8 ✓ 0,2
4 1,0	5 1,1	9 1,2
0 2,0	7 2,1	0 2,2

r	c	e	o	z
0 1 2	0	0 1 2 ✓	0 ✓	0 1 ✓
0 1 2	1	0 1 ✓	0 1 2 ✓	0 ✓
0 1 2	2 ✓	0 1	0 1	0 1

	Even	odd	zero
1-col	2 ✓	0 ✓	1 ✓
2-col	1 ✓	2 ✓	0 ✓
3-col	1 ✓	1 ✓	1 ✓

TC

File Edit Run Compile Project Options Debug Break/watch

Line 2 Col 1 Insert Indent Tab Fill Unindent * E:2PM.C

```c
#include<stdio.h>
#include<conio.h>
void main()
{
int a[10][10],nr,rs,cs, r,c; clrscr();
printf("Enter no of rows  ");scanf("%d" ,&nr);
printf("Enter %d elements\n",nr*nr);
for(r=0;r<nr;r++)for(c=0;c<nr;c++)scanf("%d",&a[r][c]);
for(r=0;r<nr;r++)
{for(rs=cs=c=0;c<nr;c++)
{rs+=a[r][c]; cs+=a[c][r];}
a[r][c]=rs; a[c][r]=cs;
}
for(r=0;r<=nr;r++)
{
for(c=0;c<=nr;c++)if(!(r==nr && c==nr))printf("%4d",a[r][c]);
printf("\n");
}
getch();
}
```

3:01 PM 25-Sep-24

TC

```
Enter no of rows  2
Enter 4 elements
1 2
3 4
   1   2   3
   3   4   7
   4   6
```

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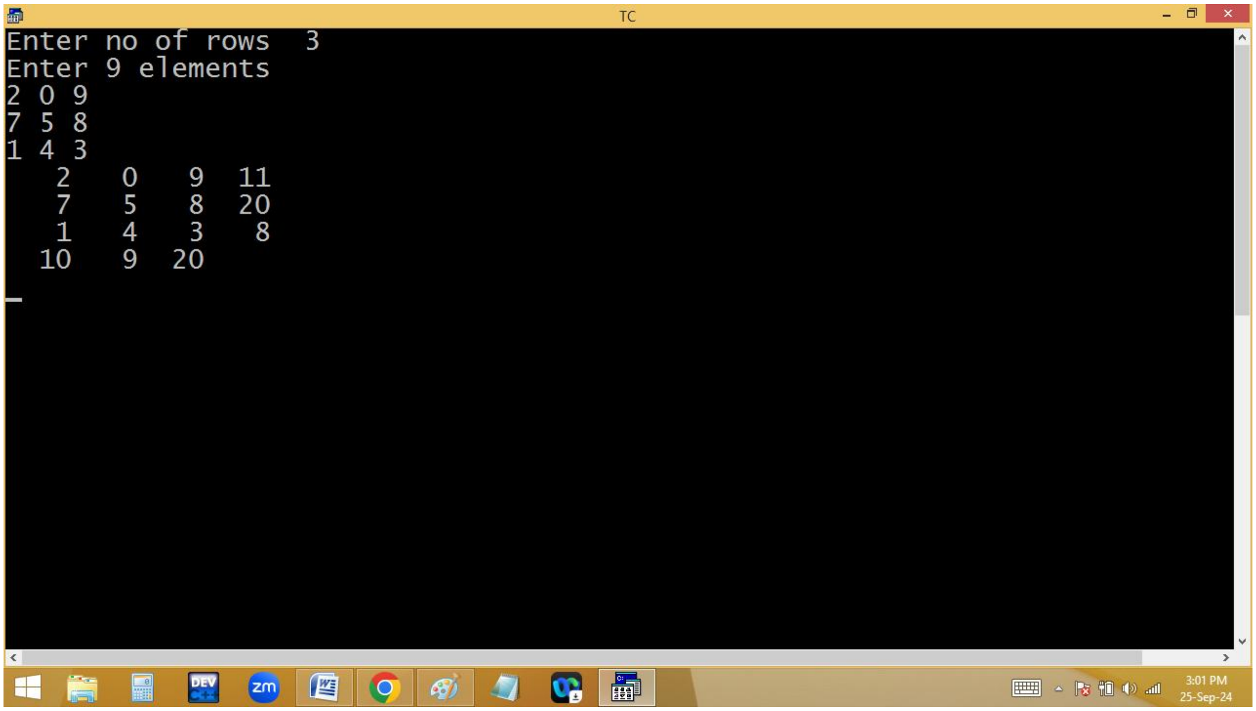
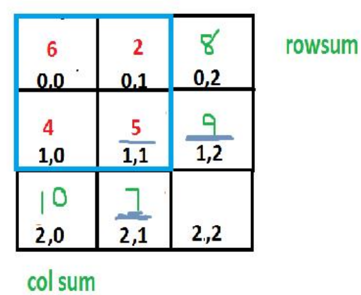


2 1 3
2 3 5
4 4

```
for( r=0; r<2; r++)
{
for( rs=cs=c=0; c<2;c++)
{
rs=rs+a[r][c];
cs=cs+a[c][r];
}
a[r][c]=rs;
a[c][r]=cs;
}
```

r	c	rs	cs
0	0 1 2	0+6+2= 8	0+6+4=10
1	0 1 2	0+4+5=9	0+2+5=7



Read n stu id, name, 6 sub marks and find the tot,avg, result.

```
#include<stdio.h>

#include<conio.h>

void  dummy(float a) {float *p = &a;}

void main()

{
```

```
float a[10][10]={0}; int  r,c,n; char  name[10][20];

clrscr();

printf("Enter no of students ");scanf("%d",&n);

for(r=0;r<n;r++)

{

printf("Enter %d stu id ",r+1);

scanf("%f",&a[r][0]);

printf("Enter name ");flushall();

gets(name[r]);

printf("Enter 6 sub marks ");

for(c=1;c<=6;c++)

{scanf("%f",&a[r][c]);a[r][7]+=a[r][c];if(a[r][c]<35)a[r][9]=-1;}

a[r][8]=a[r][7]/6.0;

}

puts("Id\tName\tTot\tAvg\tGrade");

puts("------------------------------------------------------------");

for(r=0;r<n;r++)

{

printf("%.0f\t%s\t%.0f\t%.2f\t",a[r][0],name[r],a[r][7],a[r][8]);

if(a[r][9]==-1)puts("Failed");
```

else if(a[r][8]>=75)puts("Distinction");

else if(a[r][8]>=60)puts("1st Class");

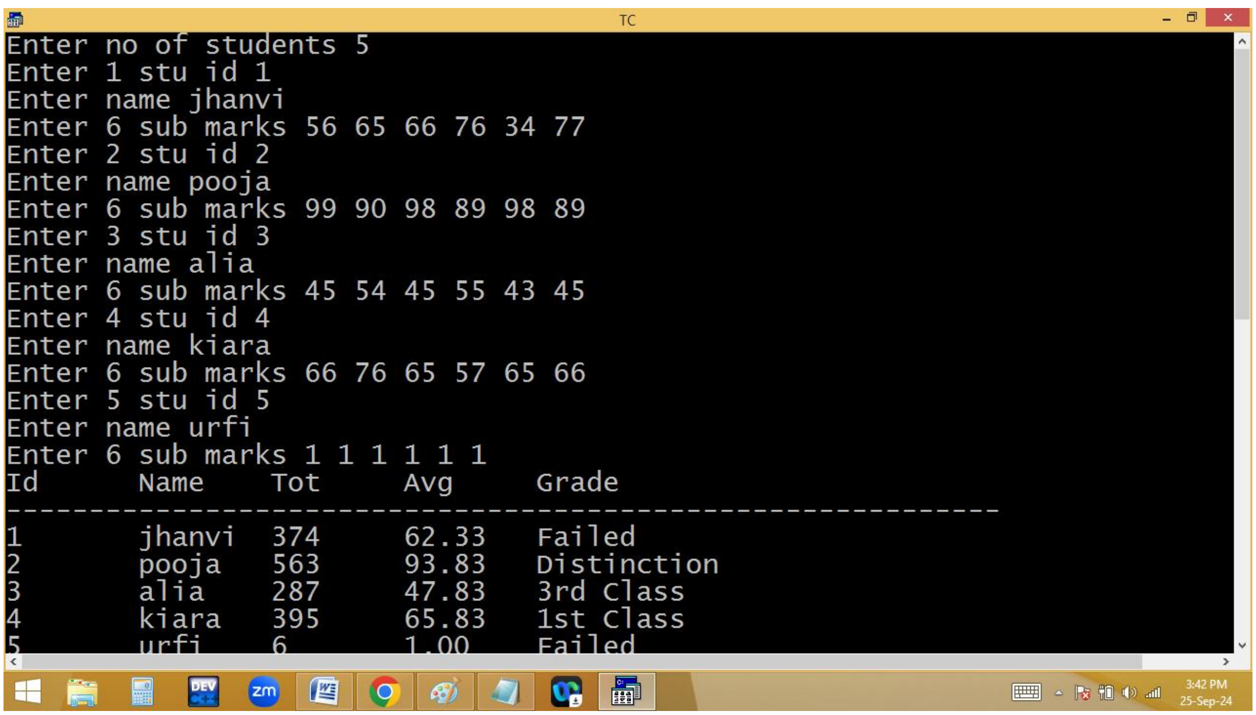
else if(a[r][8]>=50)puts("2nd Class");

else puts("3rd Class");

}

getch();

}



```
Enter no of students 5
Enter 1 stu id 1
Enter name jhanvi
Enter 6 sub marks 56 65 66 76 34 77
Enter 2 stu id 2
Enter name pooja
Enter 6 sub marks 99 90 98 89 98 89
Enter 3 stu id 3
Enter name alia
Enter 6 sub marks 45 54 45 55 43 45
Enter 4 stu id 4
Enter name kiara
Enter 6 sub marks 66 76 65 57 65 66
Enter 5 stu id 5
Enter name urfi
Enter 6 sub marks 1 1 1 1 1 1
Id      Name    Tot     Avg     Grade
------------------------------------------------------------
1       jhanvi  374     62.33   Failed
2       pooja   563     93.83   Distinction
3       alia    287     47.83   3rd Class
4       kiara   395     65.83   1st Class
5       urfi    6       1.00    Failed
```

id	tel	eng	hin	mat	sci	soc	tot	avg	result
1	x	x	x	x	x	x	x	x	0
0	1	2	3	4	5	6	7	8	9

name[0]	
name[1]	
name[2]	
name[n]	

Home work:

Fractions of matrix:

a

5.5 0,0	3.3 0,1
9.1 1,0	8.7 1,1

/

b

3.9 0,0	4.4 0,1
6.6 1,0	4.9 1,1

Matrix multiplication:

a

1 0,0	2 0,1
3 1,0	4 1,1

*

b

5 0,0	6 0,1
7 1,0	8 1,1